

CLOUD COMPUTING - TY CSE AIML

Question Bank

Unit-I: Introduction to Cloud Computing

1. Explain the evolution of cloud computing, highlighting the role of time-sharing, grid computing, and virtualization.
2. Compare cloud computing architecture with traditional computing architecture.
3. Discuss the three service models of cloud computing (IaaS, PaaS, SaaS) with suitable examples.
4. Differentiate between public, private, hybrid, and community cloud deployment models.
5. Describe the key properties and characteristics of cloud computing such as scalability, elasticity, and availability.

Unit-II: Virtualization

1. Explain the concept of virtualization. Discuss different approaches to virtualization with examples.
2. What is a hypervisor? Differentiate between Type-1 and Type-2 hypervisors with suitable examples.
3. Describe the role of Amazon Elastic Compute Cloud (EC2) as a computing service in the cloud.
4. Explain memory virtualization techniques such as ballooning, deduplication, and sharing.
5. Compare pre-copy and post-copy virtual machine migration techniques. How do they ensure system availability?

Unit-III: Cloud Services

1. Explain the role of **Service Oriented Architecture (SOA)** in enabling cloud computing. How does it relate to cloud services?

2. Differentiate between **IaaS, PaaS, and SaaS** with suitable real-world examples.
3. What is **multi-tenancy** in Software as a Service (SaaS)? Explain its significance with deployment modes.
4. Describe how **containerization and orchestration** are achieved using **Docker** and **Google Kubernetes Engine (GKE)**.
5. Discuss the importance of the **GCP Console** and **Cloud SDK** in managing cloud projects and billing.

Unit–IV: Cloud Storage

1. Differentiate between **structured and unstructured storage** in the cloud with suitable examples.
2. Explain the features and use cases of **Cloud SQL** and **Cloud Spanner** as managed services.
3. Compare **NoSQL managed services**: Cloud Datastore and Cloud Bigtable.
4. Discuss the role of **OpenStack components** (NOVA, Neutron, Keystone, Cinder, Swift, Glance) in cloud storage and management.
5. What are **Data Lakes** and how does **Snowflake** support cloud-based data storage and analytics?

Unit–V: Service Management

1. What is a **Service Level Agreement (SLA)**? Explain its importance in cloud computing with suitable examples.
2. Describe the **shared responsibility security model** in the cloud. How does it ensure better security?
3. Explain the role of **Cloud IAM** in authentication and authorization. Mention best practices for managing authorization.
4. Discuss the importance of **billing and accounting in GCP**. How does cloud billing differ from traditional IT billing?
5. What is the **DevOps architecture**? How do configuration management tools like **Ansible** support cloud services?

Unit–VI: Cloud Network and Security

1. Define a **Virtual Private Cloud (VPC)**. How does it support secure networking in the cloud?
2. Explain the role of **routes and firewall rules** in managing cloud network traffic.
3. Differentiate between **VPNs, interconnecting, and direct peering** in building hybrid clouds.
4. Discuss different **load balancing options** in the cloud and their importance in scaling applications.
5. What are the key **encryption techniques and options** used in cloud security?